

MANIDEEP POTHKAN

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SUMMARY

Full-stack developer experienced in building web applications with React, FastAPI, and Node.js. Completed a 6-month internship at ADVI Group of Companies, developing Python modules for autonomous vehicle simulation systems within cross-functional Agile teams. Comfortable across the stack — RESTful API design, relational and NoSQL database schema design, and writing clean, version-controlled, test-covered code. Built several independent projects exploring blockchain, trading logic, and audit-trail systems.

EDUCATION

Bachelor of Technology – Computer Science and Business Systems | Dec 2021 – Sep 2025
Vignana Bharathi Institute of Technology, Hyderabad · CGPA: 7.0

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL
Frontend: React.js, HTML, CSS, Tailwind CSS, Material-UI
Backend: FastAPI, Flask, Node.js, Express.js, RESTful API Design
Databases: PostgreSQL, SQLite, MySQL, MongoDB
Tooling & Deployment: Git, Docker, Vercel, Streamlit Cloud
Blockchain: Custom Blockchain Implementation, Proof-of-Work Consensus, SHA-256 Hashing
ML & Data: scikit-learn, Pandas, NumPy, yfinance, Technical Analysis
Testing & Auth: pytest, JWT, bcrypt
Visualization: Tableau, Power BI, Plotly, Recharts, Streamlit

WORK EXPERIENCE

Python Developer Intern · ADVI Group of Companies, Hyderabad | Jan 2025 – Jun 2025

- Developed Python-based simulation scripts for autonomous vehicle systems, processing sensor data and implementing vehicle behavior models for testing self-driving algorithms
- Collaborated with senior developers in Agile sprints to translate technical requirements into modular, reusable Python code
- Maintained code quality and version control using Git, including peer code reviews and merge-conflict resolution across feature branches
- Wrote unit tests with pytest and documented API endpoints and module functionality for knowledge transfer

CERTIFICATIONS

- AWS Academy Cloud Foundations – AWS
- DSA using Java – Infosys
- Google Data Analytics Professional Certificate – Google
- Solutions Architecture Job Simulation – Forage
- Software Development – Datacom

PROJECTS

AI-Powered Stock Trading Platform

React.js, FastAPI, SQLite, Technical Analysis (RSI / MACD / Bollinger Bands)

- Built a full-stack trading simulator with live stock price tracking, rule-based trade signals, and portfolio management for 100+ US/Indian stocks
- Designed a technical-indicator scoring engine (RSI, MACD, Bollinger Bands) mapping composite signal strength to confidence-labeled buy/sell/hold recommendations
- Built RESTful APIs with FastAPI handling authentication, trading operations, and portfolio analytics (Sharpe ratio, P/L tracking)
- Built a responsive, dark-mode React frontend with lazy-loaded components

VerifyChain

Python, FastAPI, SQLite/PostgreSQL, React, TypeScript, Material-UI

- Built a document verification platform on a custom blockchain with Proof-of-Work consensus and SHA-256 hashing to make tampering detectable
- Implemented core chain logic (block creation, hash linking, PoW mining, chain validation) plus a FastAPI backend with JWT authentication and bcrypt password hashing
- Designed a document upload → hash → verify workflow so re-uploaded documents can be checked for integrity against the stored on-chain hash
- Built a React + TypeScript frontend with Redux Toolkit and Material-UI
- Organized the backend into clean domain modules (auth, documents, blockchain) as a foundation for a future microservices split

Decision Audit System

Python, FastAPI, MongoDB, REST APIs

- Built an audit system capturing decision execution traces across multiple workflow types for compliance and debugging
- Designed a flexible MongoDB schema with version history supporting multiple decision types and rollback capability
- Built REST APIs for retrieving historical decisions with human-readable explanations for compliance review
- Built a dashboard UI for browsing and searching audit records
- Containerized the application with Docker and Docker Compose

Synthetic Data Generator

Python, Streamlit, Pandas, Faker, Plotly, scikit-learn

- Built a Streamlit platform generating 100K+ realistic synthetic records for ML testing using Faker with configurable distributions and constraints
- Used scikit-learn for data preprocessing and validation to catch quality issues before export
- Built real-time Plotly visualizations for instant distribution and data-quality checks

Vidhya – Student Alumni Networking Platform

React.js, Node.js, Express.js, MySQL

- Built an alumni networking platform connecting students with mentors for career guidance and job referrals
- Built REST APIs with JWT authentication and role-based access control for students, alumni, and admin roles
- Designed a normalized MySQL schema to reduce data redundancy
- Built a responsive frontend with React and Tailwind CSS